Appl. Phys. A 67, 637–644 (1998)

Applied Physics A
Materials
Science & Processing

Interaction of atomic hydrogen with metal surfaces

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Received: 23 March 1998/Accepted: 21 August 1998

Abstract. The interaction of atomic hydrogen (deuterium) with clean and precovered nickel and aluminum single-crystal surfaces has been investigated. The kinetics has been studied by thermal desorption spectroscopy and direct monitoring of the reaction products with a multiplexed mass spectrometer. Dynamical studies in terms of the rotational and vibrational state distribution of the reaction products have been performed using REMPI spectroscopy. The initial sticking coefficient of atomic hydrogen is close to unity on nickel and about 0.6 on aluminum. Subsurface absorption can be observed on nickel but not on aluminum single-crystal surfaces. Atomic hydrogen (deuterium) interacts with pre-adsorbed deuterium (hydrogen) in form of Eley–Rideal reactions, hot-atom reactions and collision-induced desorption. The internal states of the reaction products (rotation, vibration) are highly excited.

PACS: 68.10.Jy; 68.45.Da; 82.65.Pa

The interaction of reactive atoms with solid surfaces has recently attracted considerable interest because of the technological relevance of this process. In particular the reaction of atomic hydrogen and deuterium with metal and semiconductor surfaces has become a model system for atom/surface reactions. There is now ample experimental [1–4] and theoretical [5–7] evidence that the use of reactants in atomic form opens new interaction channels, not existent for molecular reactants. The most frequently observed phenomenon is the direct abstraction of adsorbed atoms by an impinging species. In this type of reaction (Eley–Rideal reaction [8]) the impinging atom is not equilibrated to the surface and therefore the reaction product is typically found in a hyperthermal state. Besides the direct interaction between the incoming atom and the adsorbed atom the impinging atom may become temporarily trapped in a so-called hot-precursor state prior to reaction. This type of reaction is frequently called Harris–Kasemo or hot-atom reaction [9]. Another feature of atom/surface interactions is the possibility of subsurface absorption [10]. The driving force for all of these processes is the large potential energy of the incoming atom with respect to the energy of the recombined molecule (half dissociation energy). During the approach of the atom to the surface this potential energy is converted into kinetic energy, which may be high enough either to push the atom into subsurface sites or to form a molecule that can leave the surface immediately, even at low surface temperature. The available energy can be channeled partially into the internal states (vibration, rotation) and the kinetic energy of the reaction product, as well as into surface excitations (phonons, electron–hole pairs).

In this contribution I will focus on the interaction of atomic hydrogen and deuterium with metal surfaces. Experimental results for the reaction of H (D) with the transition metal nickel and the simple metal aluminum will be presented. Both the kinetics (overall reaction rates) as well as the dynamics (internal state distribution, angular distribution) will be dealt with. Special emphasis has been put on obtaining quantitative data on the sticking and reaction coefficients.

1 Experimental

The kinetics experiments have been carried out in a commercial ultrahigh vacuum chamber equipped with the necessary analytical and preparative devices (Auger spectrometer, LEED optics, ion sputter gun). For atomic-hydrogen dosing a highly effective source of the Bertel type [11] has been used. This doser has been characterized in terms of the degree of dissociation and the angular distribution of the effusing particles [12]. Therefore the reaction coefficients (sticking coefficient, abstraction coefficient) can be determined quantitatively. With a tiltable sample holder angle-resolved reaction and sticking coefficients can be determined. A directional flux detector has been applied to acquire angle-resolved thermal desorption spectra [10].

The experiments concerning the vibrational and rotational state distribution of Eley–Rideal reaction products have been performed in an ultrahigh vacuum chamber equipped with a REMPI (resonance-enhanced multiphoton ionization) spectrometer. A Nd-YAG laser-pumped dye laser has been used to produce tunable laser light in the UV regime (200 nm–207 nm). With this radiation hydrogen (deuterium) molecules in distinct rotational and vibrational states can be excited resonantly from the electronic ground state into the electronically excited E,F state and subsequently ionized. The REMPI spectra obtained have been calibrated with the help of REMPI spectra of a Maxwellian beam produced in a Knudsen source [13].

2 Results and discussion

2.1 The initial sticking coefficient of atomic hydrogen

For many years it has been assumed that the sticking coefficient of atomic hydrogen on metal surfaces is unity. But when Bertel et al. [14] determined an initial sticking coefficient of 0.18 for hydrogen on Cu(110), experimental and theoretical work was triggered to elucidate this subject in more detail. We have performed sticking coefficient measurements on nickel and aluminum single-crystal surfaces. These systems were chosen because the adsorption behavior of these surfaces for *molecular hydrogen* is very different. On the nickel surfaces the initial sticking coefficient for a molecular beam with normal incidence is between 0.01 and 0.5, depending on the surface structure and the beam energy [15]. On the aluminum surfaces the initial sticking coefficient for molecular hydrogen is negligible at beam energies corresponding to room temperature and increases to about 4×10^{-4} only at beam energies of 2000 K [16]. There is also no significant dependence on the surface structure.

The beam of atomic hydrogen is produced in the Bertel source by thermal dissociation at about 2000 K. Due to experimental restrictions we cannot change the mean translational energy of the atomic hydrogen beam. In Fig. 1 the uptake curves for atomic hydrogen ($E_{\mathrm{trans}} \approx 2000$ K) and molecular hydrogen ($E_{\mathrm{trans}} \approx 300$ K) on Ni(110) are compared. From the initial slope we determine an initial sticking coefficient of 0.38 for molecular hydrogen but 0.9 for atomic hydrogen. The saturation coverage amounts to 1.5 monolayer for adsorption of molecular hydrogen. With atomic hydrogen a coverage beyond the molecular saturation can be reached due to subsurface absorption, as will be discussed in the next paragraph. A similar behavior is found for hydrogen adsorption on Ni(111). The initial sticking coefficient for hydrogen atoms again is close to unity and a coverage larger than 1 monolayer (molecular saturation) can be reached due to subsurface absorption [17]. The sticking coefficient for a molecular beam of 300 K in contrast is only 0.11. Atomic hydrogen adsorption on aluminum single-crystal surfaces shows a somewhat different adsorption behavior. The initial sticking coefficient is significantly smaller than unity on Al(100) and Al(111) and no subsurface absorption can be observed [1]. A comparison of the initial sticking coefficients for atomic hydrogen and molecular hydrogen on nickel and aluminum single-crystal surfaces can be seen in Table 1.

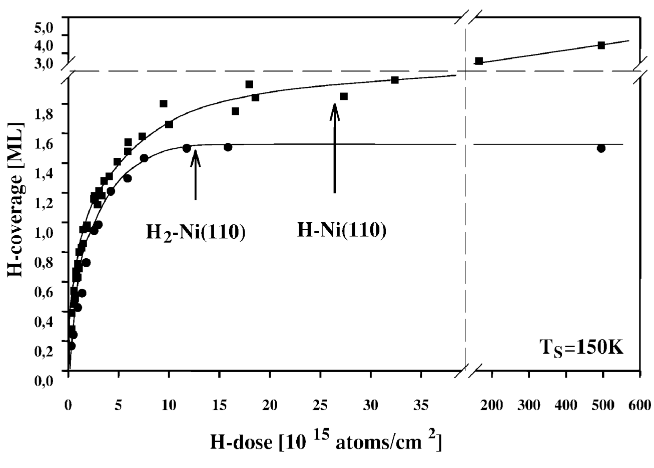

Fig. 1. Uptake curves for atomic and molecular hydrogen on Ni(110) at a surface temperature of 150 K

Table 1. Initial sticking coefficients S_0 for atomic and molecular hydrogen beams at normal incidence on various metal surface. Within the experimental error the initial sticking coefficient for atomic deuterium is the same

Metal surface	S_0(mol., 300 K)	S_0(mol., 2000 K)	S_0(atomic, 2000 K)
Ni(110)	0.38	0.35	0.9 ± 0.1
Ni(111)	0.11	0.38	1.0 ± 0.1
Al(100)	$<10^{-4}$	4×10^{-4}	0.6 ± 0.1
Al(111)	$<10^{-4}$	4×10^{-4}	0.6 ± 0.1
Au(poly)	–	–	0.5 ± 0.1

The initial sticking coefficients mentioned so far have been obtained for hydrogen atoms with normal incidence. Measurements of the angular variation of the sticking coefficient should give us additional information with regard to the specific adsorption process involved. It is well known that for *molecular hydrogen* the angular distribution of the sticking coefficient depends strongly on the particular chemical composition and geometric structure of the surface. On Ni(111) the angular variation of the sticking coefficient is forward focused, whereas on Ni(110) the sticking coefficient is nearly independent of the incidence angle [18]. On the aluminum single-crystal surfaces the angle dependence of the sticking coefficient for hydrogen molecules is also strongly forward focused [16]. The change of the sticking coefficient with incidence angle for atomic hydrogen on Ni(111) and Al(111) is plotted in Fig. 2a and Fig. 2b. In the case of Ni(111) the initial sticking coefficient is independent on the angle of incidence within the experimental error. On the Al(111) surface, for which the sticking coefficient is 0.6 at normal incidence, the sticking coefficient even increases with increasing angle of incidence. In both cases the [110] direction has been chosen as tilt axis.

The question now arises as to the mechanism of the energy dissipation responsible for the sticking of atomic hydrogen on metal surfaces. Typically the impinging atom is accelerated in a steep adsorption potential of about 2–3 eV. The kinetic energy gained in the adsorption well can be released either into phonon excitation or into electron–hole pair excitation. Unfortunately, there is disagreement in the literature as to the relative importance of these mechanisms for the H–metal system. It is generally assumed that energy dissipation by phonon excitation should play a minor role due to the large mass mismatch [19]. Although the contribution of electronic excitations in the accommodation of hydrogen atoms is discussed quite controversially [20], it is safe to assume that the energy release per single impact can only be small. Therefore the most probable explanation for the high sticking coefficient is that the energy release proceeds through a large number of interaction steps. During the first encounter the incident atom can transfer

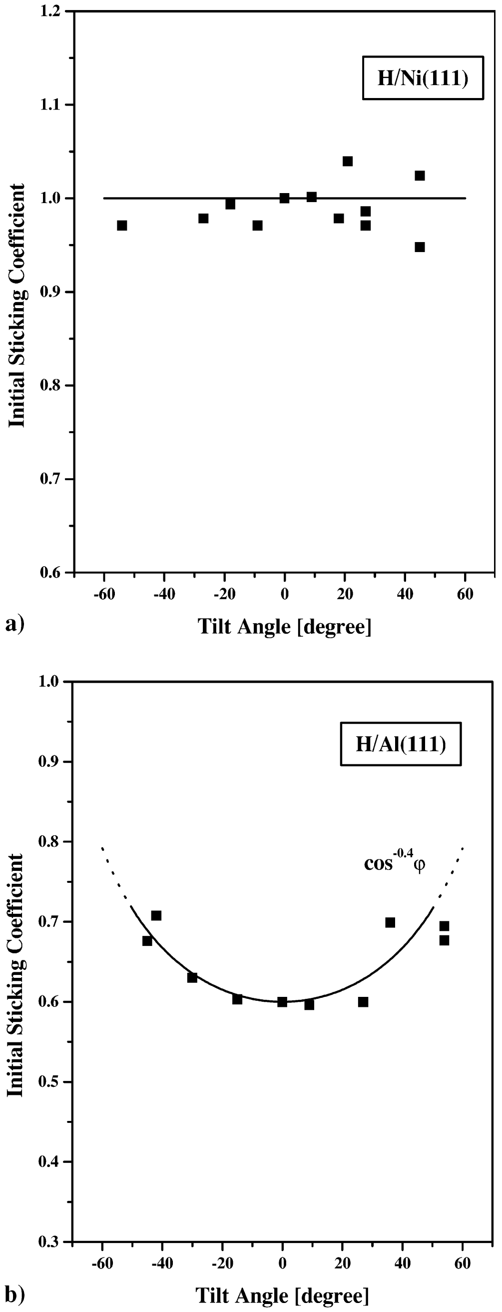

Fig. 2a,b. Angular variation of the initial sticking coefficient for atomic hydrogen on Ni(111) (**a**) and on Al(111) (**b**). Surface temperature: 130 K, tilt axis: [110] direction

part of its kinetic energy normal to the surface into motion parallel to the surface (normal-to-parallel momentum transfer). If this energy transfer is large enough the particle becomes bound to the surface in the form of a hot atom. During diffusion across the surface the atom can gradually lose energy and become eventually accommodated. The increase of the initial sticking coefficient with increasing incidence angle for H on Al(111) would point towards such a mechanism.

The difference of the sticking coefficients for nickel (close to unity) and aluminum surfaces (≈ 0.6) can be explained by the different potential well depth (binding energy: difference between the energy of the free atom and the energy of the adsorbed atom). The binding energy can be calculated from the potential energy of the free hydrogen atom ($E_{\mathrm{pot}} = \frac{1}{2}E_{\mathrm{diss}} = 2.25\,\mathrm{eV}$), the desorption energy E_{des} and the activation barrier for adsorption E_{act} [1]: $E_{\mathrm{bind}} = E_{\mathrm{pot}} + E_{\mathrm{des}} - E_{\mathrm{act}}$. The desorption energy as obtained from thermal desorption spectroscopy is about 0.37 eV/H-atom on aluminum [21] and about 0.48 eV/H-atom on nickel [22]. The activation barrier for adsorption as obtained from molecular beam experiments is smaller than 0.05 eV/H-atom on nickel [15], but about 0.5 eV/H-atom on the aluminum surfaces [16]. Therefore the binding energy for a H atom can be calculated to be 2.7 eV on nickel and 2.1 eV on aluminum, respectively. This difference in the binding energy could be responsible for the different sticking coefficients.

We have looked for an isotope effect in adsorption by replacing hydrogen atoms by deuterium atoms. For all of the systems investigated we were unable to find a significant difference in the initial sticking coefficient for H and D atoms. In particular for the system H (D)–nickel, in which the sticking coefficient is close to unity, one would not expect a difference between H and D. But also in the case of aluminum the sticking coefficients were the same within the experimental error. Considering the hot-atom adsorption mechanism it is obvious that the isotope dependence of the individual accommodation processes will be wiped out.

The coverage dependence of the sticking coefficient for hydrogen atoms is quite different compared with the coverage dependence of hydrogen molecules. For molecular adsorption the sticking coefficient always approaches zero at saturation coverage. In the case of H on nickel, subsurface absorption yields a non-negligible net sticking coefficient at surface saturation. We determined an absorption coefficient of about 1×10^{-2} for a total coverage (surface + subsurface coverage) of about 1.5 monolayers. The absorption coefficient slightly decreases with increasing subsurface coverage. Since on the aluminum surface no subsurface absorption has been observed the net sticking coefficient for atoms approaches zero at saturation coverage. On the Al(100) surface the saturation coverage is 1.6 monolayers. However, isotope exchange experiments reveal a considerable sticking rate even at saturation coverage. This is due to the fact that impinging hydrogen atoms abstract adsorbed hydrogen atoms. These empty surface sites are immediately occupied by other impinging hydrogen atoms. The effective sticking coefficients on the saturated nickel and aluminum surfaces are between 0.2 and 0.4.

2.2 Absorption of atomic hydrogen in subsurface sites

Subsurface absorption of atomic hydrogen has been frequently observed on transition-metal surfaces. This phenomenon can be easily explained within a simple one-dimensional picture. Due to the rather large potential energy of the free atom with respect to the energy of the absorbed atom it may gain enough kinetic energy in the interaction region to surmount the activation barrier for absorption. Subsurface absorption has been observed for hydrogen on Ni(111) [10, 23], Ni(100) [24], Ni(110) [25], and Cu(110) [14]. Hydrogen atoms adsorbed on a surface as a result of dissociative adsorption of molecular hydrogen usually have not enough energy to surmount this barrier.

Subsurface hydrogen typically manifests itself in thermal desorption spectra in the form of additional desorption peaks at temperatures lower than the surface recombination peaks. In Fig. 3 a series of desorption spectra following atomic deuterium adsorption on Ni(111) is depicted. Curve a in this figure represents the well-known saturation desorption spectrum obtained from molecular deuterium adsorption (β_1 and β_2 state) [26]. Sufficient exposure with atomic hydrogen leads to additional desorption peaks at lower temperatures labeled γ_1, γ_2, and α. Since we cannot saturate these new peaks we attribute these peaks to desorption from subsurface sites. This is in agreement with the work by Ceyer et al. [23] in which from HREELS experiments the same conclusion has been drawn.

The desorption spectra in Fig. 3 were obtained by detecting the deuterium signal with an out-of-line mass spectrometer in the vacuum chamber and therefore have to be regarded as angle-integrated spectra. In Fig. 4 we present angle-resolved desorption spectra for hydrogen obtained with the directional flux detector for 0°, 30°, and 57°. The spectrum labeled 2π represents the particle density increase in the detector due to molecules impinging on the capillary array isotropically. This contribution has to be subtracted from all the spectra obtained with the sample in front of the detector. A quick inspection of the change of the individual desorption peaks with desorption angle reveals the totally different angular distribution for desorption from surface and subsurface sites. Evaluation of the spectra yields an angular flux distribution in the form $D(\Phi) = D(0°)\cos^{4.5}\Phi$ for the surface states β_1 and β_2, in good agreement with previous work [27]. The strongly forward focused angular distribution is a result of the activation barrier for adsorption existing for the H_2–Ni(111) system. The subsurface desorption peaks on the other hand decrease only slightly with increasing angle

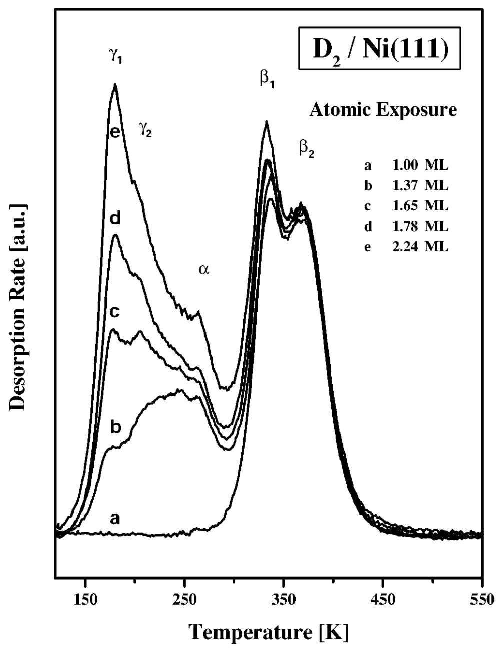

Fig. 3. Angle-integrated desorption spectra for deuterium from Ni(111). Curve a: dosing with molecular deuterium up to saturation, curves b–e: increased dosing with atomic deuterium. Adsorption temperature: 110 K, heating rate: 3 K/s

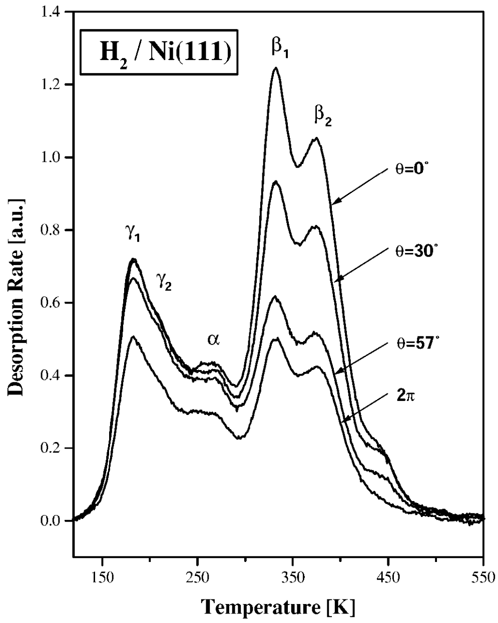

Fig. 4. Angle-resolved thermal desorption spectra for hydrogen from Ni(111) at a constant coverage of 2 monolayers. The 2π signal represents the background signal in the detector and has to be subtracted from the curves at various detection angles

and the corresponding desorption flux can be described by $D(\Phi) = D(0°)\cos^{0.6}\Phi$. This is definitely a surprising result and we can only speculate as to the reason for this peculiar behavior. One has to realize that desorption of subsurface hydrogen proceeds by recombination of a resurfacing atom with a surface atom from the saturated surface layer. The nascent hole on the surface is immediately refilled by another resurfacing atom. Throughout subsurface desorption the surface is covered by a hydrogen saturation layer. This will of course result in a potential energy surface quite different from that of the clean or partially covered surface. Generally speaking, the smaller the activation barrier for adsorption the less forward focused is the angular distribution for adsorption and desorption [28]. The lack of an activation barrier results in a cosine distribution. An angular distribution even broader than cosine is usually explained in terms of a precursor state. For our particular case ($\cos^{0.6}\Phi$) this would mean that the nascent molecule is temporarily trapped in front of the hydrogen-saturated surface prior to desorption. Another explanation could be that dynamic effects in the recombination process govern the desorption distribution. In fact theoretical considerations by Baer et al. [29] have indicated that the resurfacing atoms recombine with nearby adsorbed atoms rather than with atoms on top of the resurfacing atom. This type of recombination would result in increased parallel momentum of the nascent molecule and therefore preferentially in off-normal desorption.

2.3 *Reaction kinetics of atomic hydrogen (deuterium) with adsorbed deuterium (hydrogen)*

The most frequently observed feature of atomic hydrogen interaction with surfaces is the abstraction of previously adsorbed particles. This is usually called an Eley–Rideal reaction. Although this type of interaction has been postulated

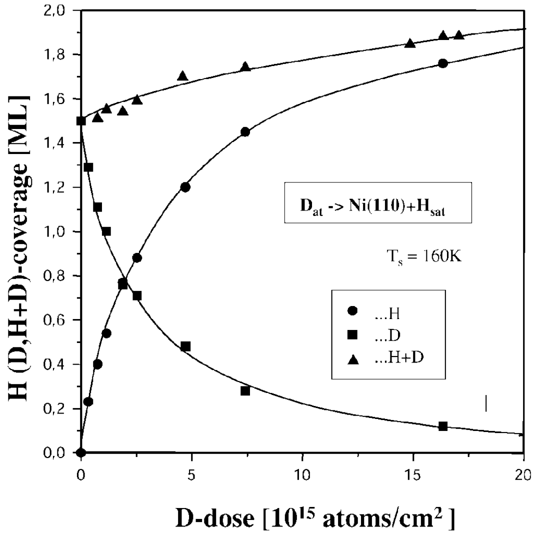

Fig. 5. Uptake of deuterium on, and removal of hydrogen from, an initially hydrogen-saturated Ni(110) surface at 160 K, exposed to a beam of atomic deuterium. Data plotted as a function of atomic-deuterium exposure

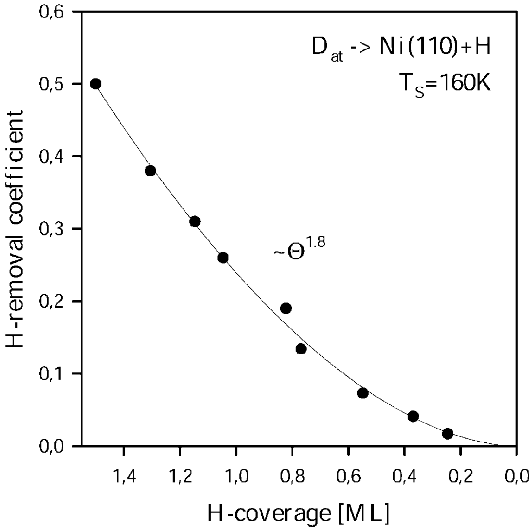

Fig. 6. Hydrogen removal per impinging deuterium atom on Ni(110) as a function of the remaining hydrogen coverage. Please note that in this case the removed hydrogen is replaced by deuterium, therefore the total surface coverage is always about 1.5 monolayers (H + D)

a long time ago [8], it has been only recently that clear experimental evidence for this process could be found [30, 31]. In particular the kinetics of such an interaction is of interest since it will give us information on the individual steps involved in the reaction. We have performed kinetic experiments with H (D) on D (H) precovered Ni(110) and Al(100) surfaces. If we dose a hydrogen-saturated Ni(110) surface for some time with atomic deuterium the surface will always remain saturated, but part of the adsorbed hydrogen has been replaced by deuterium. This can be detected by mass-multiplexed thermal desorption spectrometry, properly counting the individual atoms in the mass signals $m = 2,\ 3$, and 4. In Fig. 5 the change of the H and D coverage, as well as the total coverage (H + D) as a function of the deuterium dose is shown. The slight increase of the total coverage stems from subsurface absorption. From the initial slope of the decreasing hydrogen coverage versus exposure we calculate a removal coefficient of 0.5 H/D. The "initial" sticking coefficient for D on the hydrogen saturated surface is slightly above 0.5 due to the additional subsurface absorption. The question now arises if the removal of the adsorbed species can be described by a pure Eley–Rideal mechanism, which is characterized by a linear coverage-dependence of the reaction. Evaluation of the data of Fig. 5 (plotting the derivative of the abstraction curve versus coverage) yields a coverage dependence of the removal coefficient according to $\Theta^{1.8}$ (Fig. 6). This nonlinear coverage-dependence of the abstraction rate suggests that additional reaction mechanisms have to be involved.

To gain more insight into the reaction mechanisms we have directly monitored the reaction products originating during dosing of a deuterium-saturated Ni(110) surface with atomic hydrogen by a multiplexed mass spectrometer. Interestingly, we not only detect the HD molecules expected, but also plain D_2. It is for the first time that such a collision-induced associative desorption has been observed [17]. In Fig. 7 both the HD abstraction coefficient as well as the D_2 desorption coefficient have been plotted. Starting with the deuterium-saturated surface ($\Theta = 1.4$ monolayers) the initial abstraction coefficient is about 0.4 HD/H and the initial collision-induced desorption coefficient is 0.04 D_2/H. The decrease of the coefficients with coverage (defining the reaction order) is quite different for abstraction and collision-induced desorption: whereas the abstraction coefficient changes roughly with $\Theta^{1.3}$, the collision-induced desorption coefficient follows a second-order reaction. The collision-induced associative desorption seems to be a quite general phenomenon and has very recently also been observed by Küppers et al. [32] for Pt(111) and by Boh et al. [1] for Al(100). Classical trajectory calculations by Kratzer [7] demonstrated the existence of a removal process in the form of collision-induced desorption with a branching ratio into this reaction channel in the order of a few percent. The mechanism involved in this reaction can be explained in the following way: Some of the impinging particles can, depend-

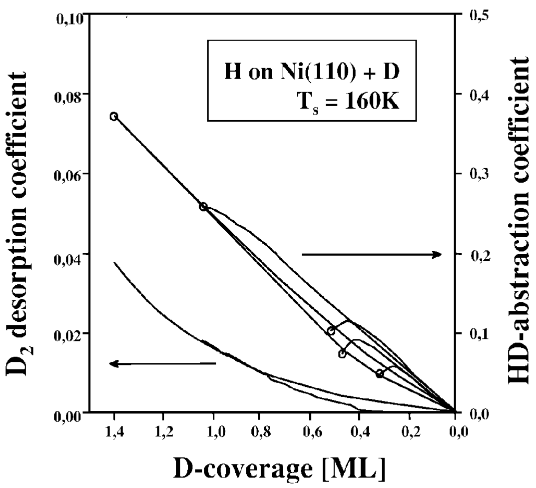

Fig. 7. HD-abstraction coefficient and D_2 desorption coefficient for atomic hydrogen impinging on a deuterium-covered Ni(110) surface at 160 K, plotted as a function of the remaining deuterium coverage. The parameter for the different curves is the different initial deuterium coverage

ing on the impact parameter, transfer a considerable amount of energy parallel to the surface to an adjacent atom. This produces a “hot-atom” [2] which then can surmount the diffusion barrier and react with another adsorbate. In a recent paper [1] we have proposed a somewhat different explanation for the collision-induced desorption process. Instead of the single-collision process we rather prefer a collective process: A hydrogen atom impinging on a fully covered layer squeezes into this layer producing a local supersaturation state. The increased surface energy can now be lowered by the recombinative desorption of two adsorbed atoms in form of a second-order (Langmuir–Hinshelwood) reaction.

The reaction order of abstraction ($\approx \Theta^{1.3}$) suggests that even this process cannot be described by a pure Eley–Rideal mechanism, but has to involve a hot-precursor or Harris–Kasemo mechanism. However, in the case of the fully precovered surface it is not easy to determine the reaction mechanism unambiguously. As mentioned above, the initially deuterium-saturated surface remains continuously saturated during the reaction due to the immediate filling of the empty sites by impinging hydrogen atoms. Only the ratio between adsorbed hydrogen and deuterium changes. Even if the abstraction (HD formation) were governed by a hot-precursor mechanism, the abstraction coefficient would always be proportional to the amount of deuterium adsorbed, as long as the reaction probability for $H + D \Rightarrow HD$ and $H + H \Rightarrow H_2$ is the same. More information on the specific reaction steps can be obtained by starting the reaction with a partially covered surface. The result can be seen in Fig. 7 for various initial deuterium coverages. The general behavior for all curves is the following: The initial HD abstraction coefficient on the partially D-covered surface is smaller than expected from the proportionality to the coverage. Apparently a hydrogen atom impinging on top of a deuterium atom can either react with this atom to form HD or it can be steered away to an adjacent free adsorption site. During the course of dosing the initially free sites become occupied by hydrogen and therefore the branching changes in favor of the abstraction reaction. With successive dosing the HD rate of course decreases again due to the decrease of adsorbed deuterium. A similar abstraction behavior has been observed by Küppers et al. for $H + D$ on Ni(100) [24] and Pt(111) [32]. The same experiment performed with exchanged isotopes yields qualitatively the same result, but the initial increase of the HD rate is not as pronounced. The reason for this behavior might be that the heavier deuterium projectile cannot be steered away as easily from the adsorbed hydrogen atom.

A different abstraction behavior can be observed on the Al(100) surface. On the deuterium-saturated surface the abstraction proceeds in a similar fashion as in the case of Ni(110) as depicted in Fig. 8, curve a: The initial abstraction coefficient is 0.4 HD/H and the coverage dependence is slightly off-linear. However, on the partially precovered surface the abstraction features are totally different. Deuterium precoverage with 1/2 monolyer and even with 1/4 monolayer leads to initial abstraction coefficients still close to 0.4. Only for very small precoverage does the initial abstraction coefficient start to decrease. The explanation for this peculiar experimental behavior is probably a distinct hot precursor on the bare aluminum surface. In fact, a similar abstraction mechanism for the system $H + D$ on Cu(111), although much less pronounced, has been described along the same line [33].

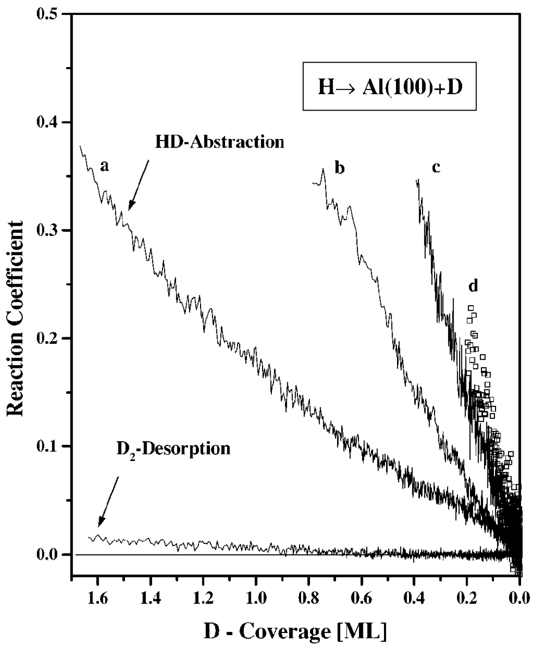

Fig. 8. HD-abstraction coefficient and D_2 desorption coefficient for atomic hydrogen impinging on a deuterium-covered Al(110) surface at 170 K, plotted as a function of the remaining deuterium coverage. The parameter for the different curves is the different initial deuterium coverage

2.4 *Internal state distribution of the reaction products*

The determination of the internal state distribution (rotation, vibration) and the translational energy of the reaction products should give us additional information on the particular reaction mechanism. The common feature of Eley–Rideal or hot-atom reactions is the rather large potential energy of the impinging atom (about half the dissociation energy) available for the reaction. It is expected that a large part of this energy is channeled into the various degrees of freedom of the reaction product. Rettner and Auerbach [4, 34] have studied the system D (H) on Cu(111). They observed the formation of fast HD molecules with a mean kinetic energy of about 1.1 eV and a forward-focused angular distribution of the reaction products. The mean rotational energy was found to be 0.4 eV and vibrational states up to $\nu = 4$ were detected. The authors came to the conclusion that for the system D-Cu(111) + H virtually all the available energy has been channeled into the three degrees of freedom of the nascent molecules and the amount of surface excitations should be negligible due to the large mass mismatch between the deuterium and copper atoms.

In this contribution equivalent experiments for the system H (D)–Ni(110) will be described. The reason for this choice is that adsorption and desorption of *molecular hydrogen* on copper and nickel single-crystal surfaces proceed quite differently. It is therefore of interest to see to what extent the electronic structure of the surface influences the reaction of atomic hydrogen (deuterium) with adsorbed H or D. The experiments have been performed in the following way [35]: A beam of atomic deuterium is aimed onto the Ni(110) surface which is held at 180 K. Under these conditions the surface is covered by a saturation deuterium layer. Therefore D_2 molecules are permanently produced by Eley–Rideal and/or hot-atom reactions as well as collision-induced desorption reaction. These molecules enter the REMPI detection chamber where they are resonantly excited and ionized by the tunable laser. We have been able to detect deuterium molecules in the vibrational states $\nu = 0, 1, 2$, and 3. This is far in excess of the thermalized equilibrium population as determined by the surface temperature of 180 K. At the same time we have detected a large number of occupied rotational states in the individual vibrational states. In Fig. 9 the rotational state distribution of the individual vibrational states is depicted in form of a Boltzmann plot. The slopes of the individual lines yield the mean rotational temperature of the D_2 molecules in the respective vibrational states. Although there is no a priori physical reason for the rotational states to obey a Boltzmann distribution after the reaction, the data points are located rather close to a straight line. It is therefore possible to characterize the rotational energy by a mean rotational temperature. We obtain rotational temperatures of 2120 K, 1840 K, 1070 K, and 520 K for the desorbing D_2 molecules in the vibrational states 0, 1, 2, and 3, respectively. This is far in excess of the surface temperature of 180 K. Determination of the mean rotational energy in the individual vibrational states ($\sum P(J) \cdot E(J)$) yields 185 meV, 133 meV, 75 meV, and 37 meV for $\nu = 0, 1, 2$, and 3, respectively. The overall mean rotational energy amounts to 150 meV. One can clearly recognize a compensation effect between the rotational and vibrational states; the higher the vibrational energy the lower the rotational energy. This result is comparable with the findings by Rettner and Auerbach [34]. Theoretical work carried out by Persson and Jackson [36] also predicts a compensation effect between rotational and vibrational degrees of freedom.

The population of the individual vibrational states in the form of a Boltzmann plot is shown in Fig. 10. The mean vibrational energy ($\sum P(\nu) \cdot E(\nu)$) is 220 meV, the zero point energy not included. There is, of course, again no need for the data to follow a Boltzmann distribution since the population is not determined by an equilibrium situation. Just to get a qualitative impression of the population distribution we

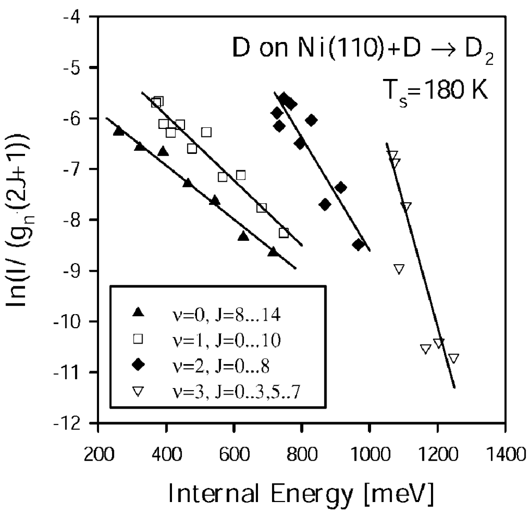

Fig. 9. Boltzmann plots for the rotational state distribution of deuterium molecules resulting from D-atom interaction with a deuterium-covered Ni(110) surface at 180 K. The molecules observed are in vibrational states $\nu = 0-3$. The *straight lines* are least-square fits to the data points and yield rotational temperatures as given in the text

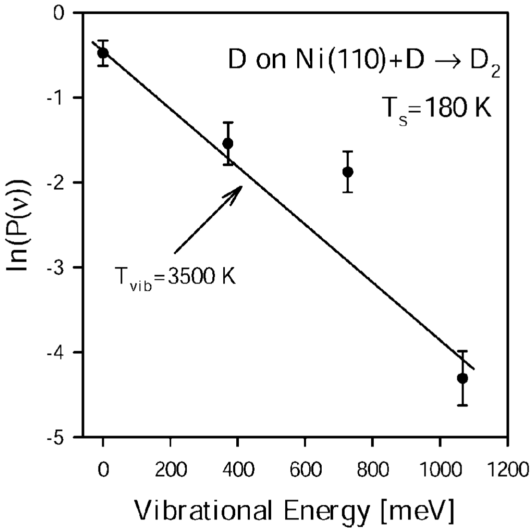

Fig. 10. Relative population of the vibrational states $\nu = 0-3$ for deuterium molecules resulting from D-atom interaction with a deuterium-covered Ni(110) surface at 180 K. The *slope of the straight line* yields a vibrational temperature of 3500 K

have approximated the data points by a straight line, which yields an equivalent vibrational temperature of 3500 K. This shows that a considerable amount of the reaction energy is channeled into the internal degrees of freedom of the reaction product. However, compared with the results for the system D-Cu(111)+H, significantly less energy is contained in all of the internal states for the system D-Ni(110)+D. One reason for the observed difference could be the contribution of the collision-induced desorption. Molecules originating from this particular reaction path will of course contain much less energy in the internal states compared with direct Eley–Rideal reaction products. Unfortunately, using the REMPI technique we cannot distinguish between molecules originating from an Eley–Rideal, Harris–Kasemo or collision-induced desorption reaction. Moreover, up to now we were not able to measure the translational energy of the state-selected molecules simultaneously, but experiments of this type are under way. Considering the mean rotational energy of 150 meV and the mean rotational energy of 220 meV it is unlikely that the rest of the available reaction energy of 2.2 eV should be contained in the translational energy (1.8 eV). We rather believe that a significant amount of the available energy has been transferred into surface excitations in the form of phonon and electron–hole pair excitation.

3 Summary

The interaction of atomic hydrogen with clean and hydrogen-covered metal surfaces reveals a variety of individual reaction processes. Hydrogen atoms can stick on the clean surface with a high probability, but not necessarily with a probability of 100%. Hydrogen atoms can even penetrate the surface to form subsurface absorption states. On a saturated surface hydrogen atoms can adsorb with a non-negligible sticking coefficient due to concurrent removal of absorbed species. This removal can take place in the form of

1. Direct abstraction (Eley–Rideal reaction)

2. Indirect reaction (hot-atom or Harris–Kasemo reaction), and
3. Collision-induced associative desorption.

The reaction products exhibit a strong non-thermal distribution of the rotational and vibrational states. It can be shown that the energy channeled into these two degrees of freedom follows a compensation effect. Molecules in higher vibrational states possess lower rotational energy and vice versa. Experiments on the kinetics and dynamics have been performed up to now on nickel, aluminum, and copper single-crystal surfaces. The results show a number of common features on the one hand but also quantitative differences on the other hand. Although we have obtained quite a good picture of the interaction of hydrogen atoms with metal surfaces so far, much more experimental and theoretical work has to be done to answer the numerous remaining questions.

Acknowledgements. I gladly acknowledge the contributions of my coworkers G. Eilmsteiner, W. Walkner, H. Boh, H. Premm, and H. Pölzl, as well as the stimulating discussions with K. Rendulic. Financial support was given by the Austrian Fonds zur Förderung der wissenschaftlichen Forschung.

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